

1. Objective

The objective of this lab is to explore frequent word patterns and co-occurrences in cleaned YouTube comments and captions data. Using the datasets prepared in Lab 2, this lab focuses on identifying words that frequently appear together, visualizing their relationships, and discovering association rules using the Apriori algorithm. The lab emphasizes understanding patterns through both manual inspection and algorithmic pattern mining, rather than formal theory.

2. Tools and Libraries Used

- **Python version:** Python 3.14.3
- **Libraries used:**
 - pandas
 - matplotlib
 - networkx
 - collections (Counter)
 - nltk (stopwords, PorterStemmer)
 - mlxtend (Apriori, association rules)
 - wordcloud

These tools were used for data handling, visualization, pattern mining, and text processing.

3. Dataset Description

Two datasets were used in this lab:

1. **cleaned_comments.csv**
Contains preprocessed YouTube comments from Lab 2. Each row includes a `cleaned_tokens` column representing tokenized words after cleaning.
2. **cleaned_captions.csv**
Contains preprocessed YouTube captions from Lab 2, also with a `cleaned_tokens` column.

During loading:

- Rows with empty or invalid token lists were removed.
- Tokens were verified to be list-like using safe parsing.
- Some very common English stopwords were later removed during co-occurrence analysis.

4. Methodology / Procedure

4.1 Setup & Data Recall

Both datasets were loaded into pandas DataFrames and validated to ensure the cleaned_tokens column existed and contained list-like data. Rows with empty token lists were removed.

The top 20 unigrams were computed using collections.Counter and visualized.

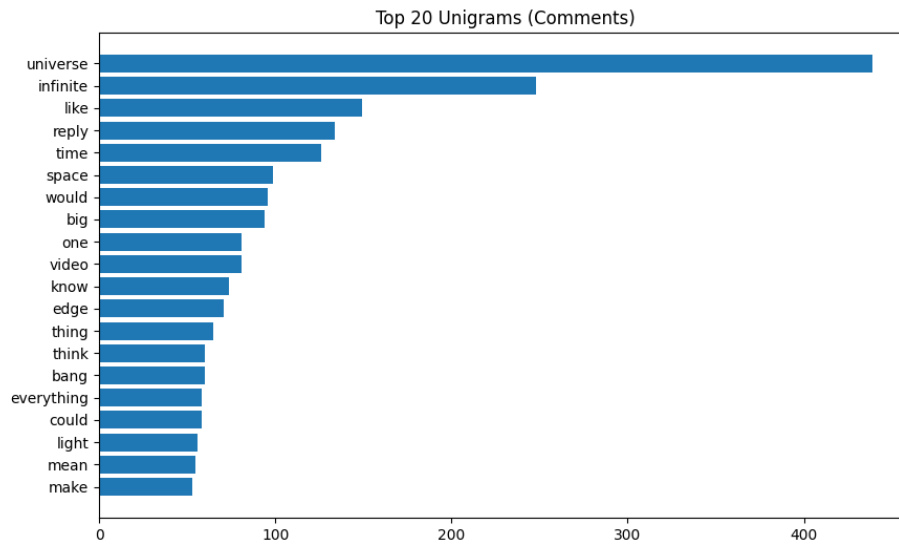


Figure 1: Top 20 Unigrams (Comments)

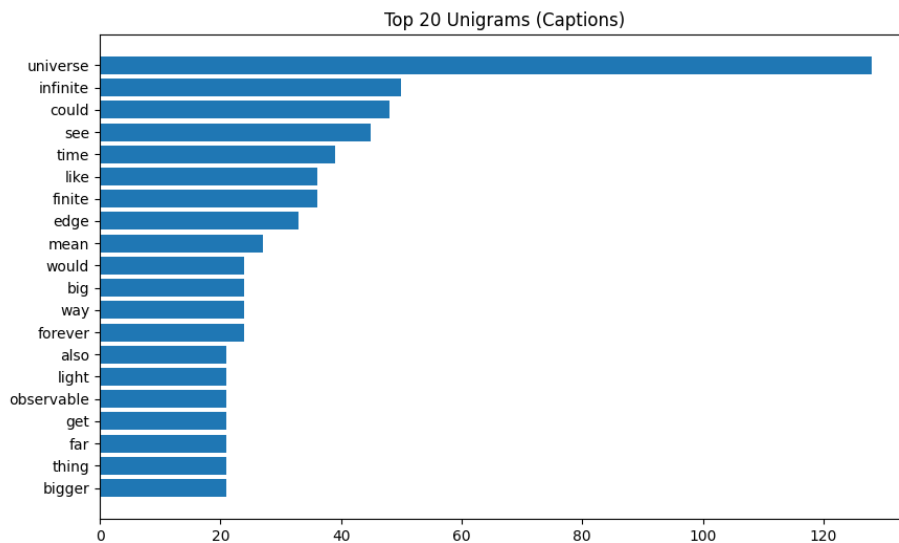


Figure 2: Top 20 Unigrams (Captions)

From both plots, words like “universe”, “infinite”, “time”, “space”, and “edge” appear frequently, indicating a strong focus on cosmology and physics-related discussions.

4.2 Transaction Construction

Each list of tokens was treated as a **transaction (basket)**.

Steps followed:

- Transactions with fewer than 3 tokens were removed.
- Duplicate tokens within each transaction were removed.
- Each transaction was assigned a unique ID (TX_0, TX_1, etc.).

A histogram of transaction lengths was plotted. Additionally, basket length statistics were computed and saved.

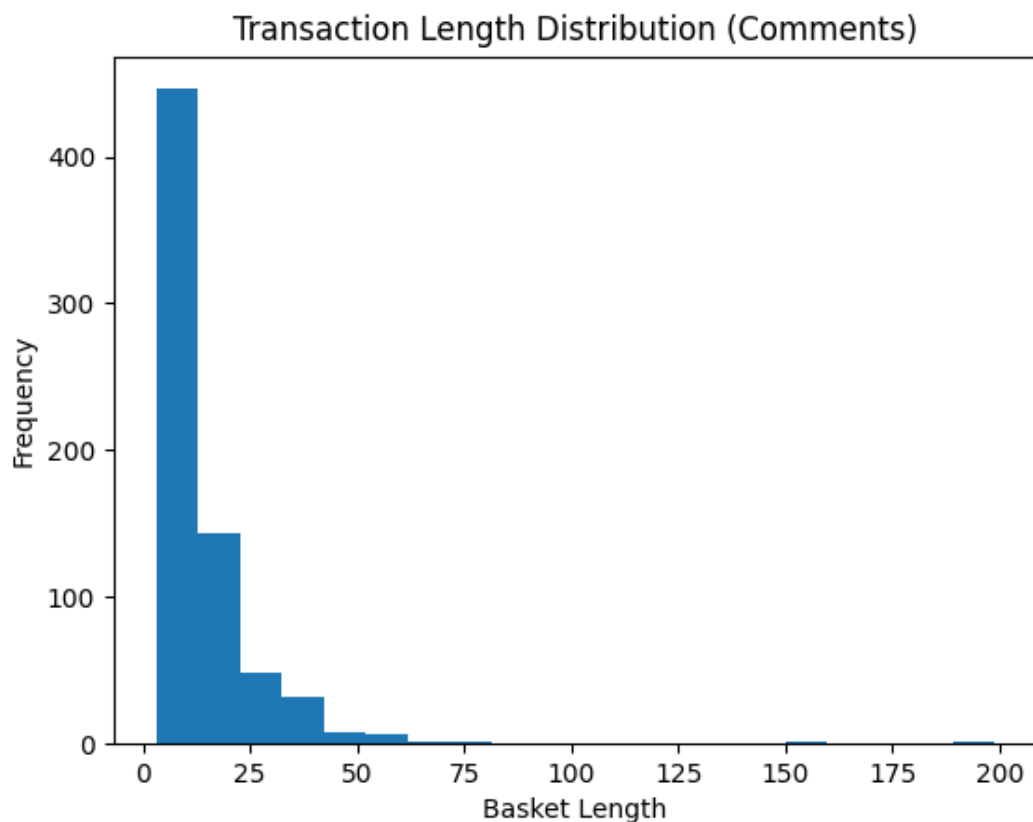


Figure 3: Transaction Length Distribution (Comments)

Table 1: Transaction Length Statistics

Metric	Value
Average Length	13.52
Minimum Length	3
Maximum Length	199

The distribution shows that most comments are short, but a few long comments create a long tail.

4.3 Manual Pattern Discovery

Pairwise word co-occurrences were counted across all transactions after removing stopwords.

Only pairs with **count** ≥ 3 were kept.

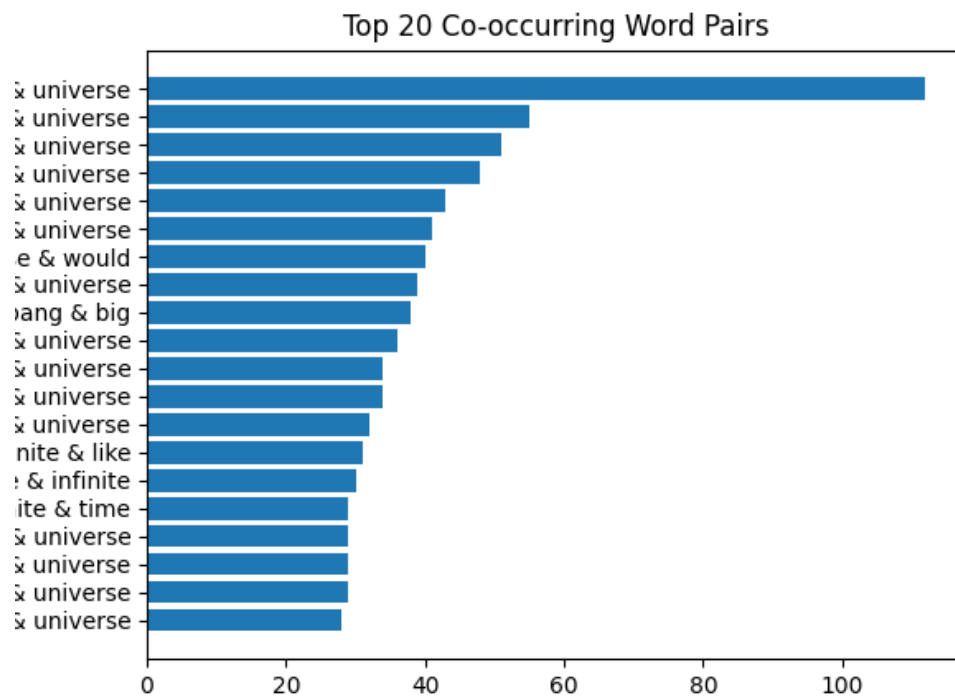


Figure 4: Top 20 Co-occurring Word Pairs

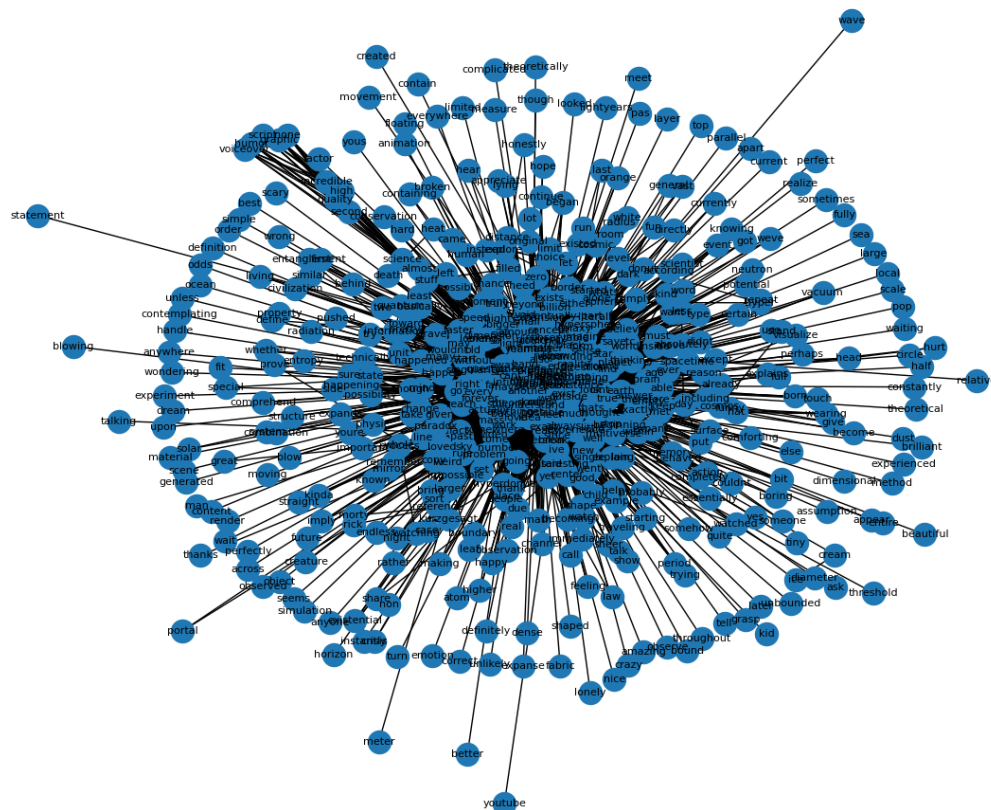


Figure 5: Co-occurrence Network Graph

Common pairs include:

- *big & universe*
- *infinite & universe*
- *space & time*

- min_support = 0.2
- min_support = 0.1

Frequent itemsets of **length 2 and 3** were retained. Association rules were generated with:

- confidence ≥ 0.6
- lift ≥ 1.2

Here, results were saved as comments_frequent_and merged datasets. Also, support vs confidence scatter plots were created.

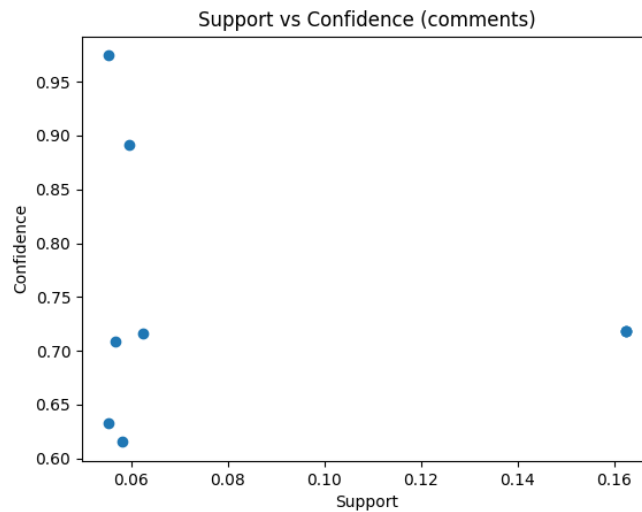


Figure 6: Support vs Confidence (Comments)

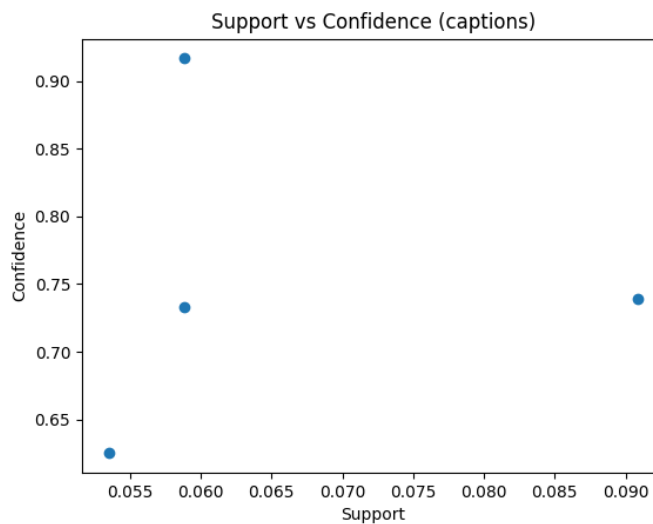


Figure 7: Support vs Confidence (Captions)

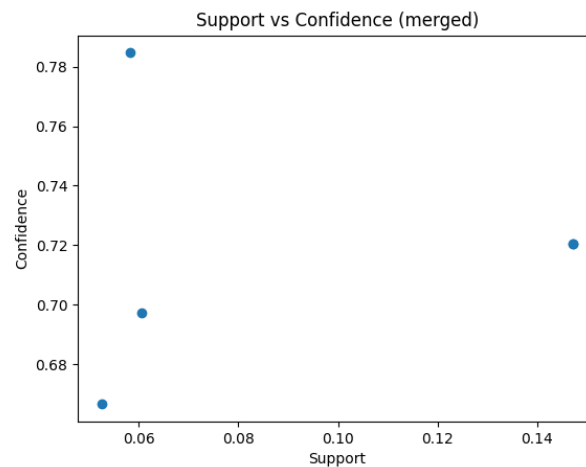


Figure 8: Support vs Confidence (Merged)

The plots show that some rules have **high confidence but low support**, meaning they are reliable but occur in fewer comments.

4.5 Exploration Using Variations

Several variations were tested:

1. **Lower support thresholds** (0.15 and 0.05)
 - Produced many additional patterns, including trivial ones.
2. **Stemming instead of lemmatization**
 - Reduced vocabulary size.
 - Increased some rule confidence.

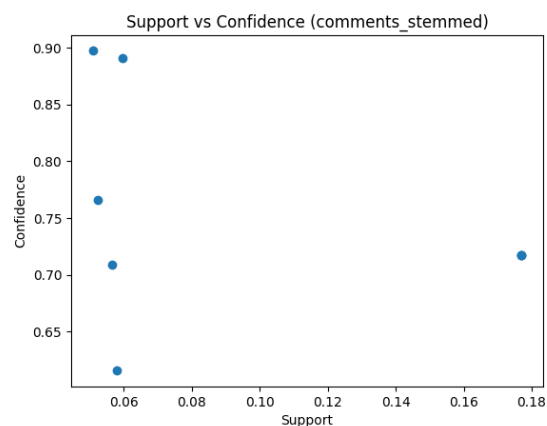


Figure 9: Support vs Confidence (Stemmed Comments)

3. Removing tokens shorter than 4 characters

- Reduced noise.
- Removed many generic words.

4. Captions-only analysis

- Patterns were fewer but more focused.

5. Merged comments + captions

- Increased support for core concepts like “universe” and “infinite”.

Comparing results across variations showed how sensitive Apriori is to preprocessing choices.

4.6 Pattern Interpretation & Visualization

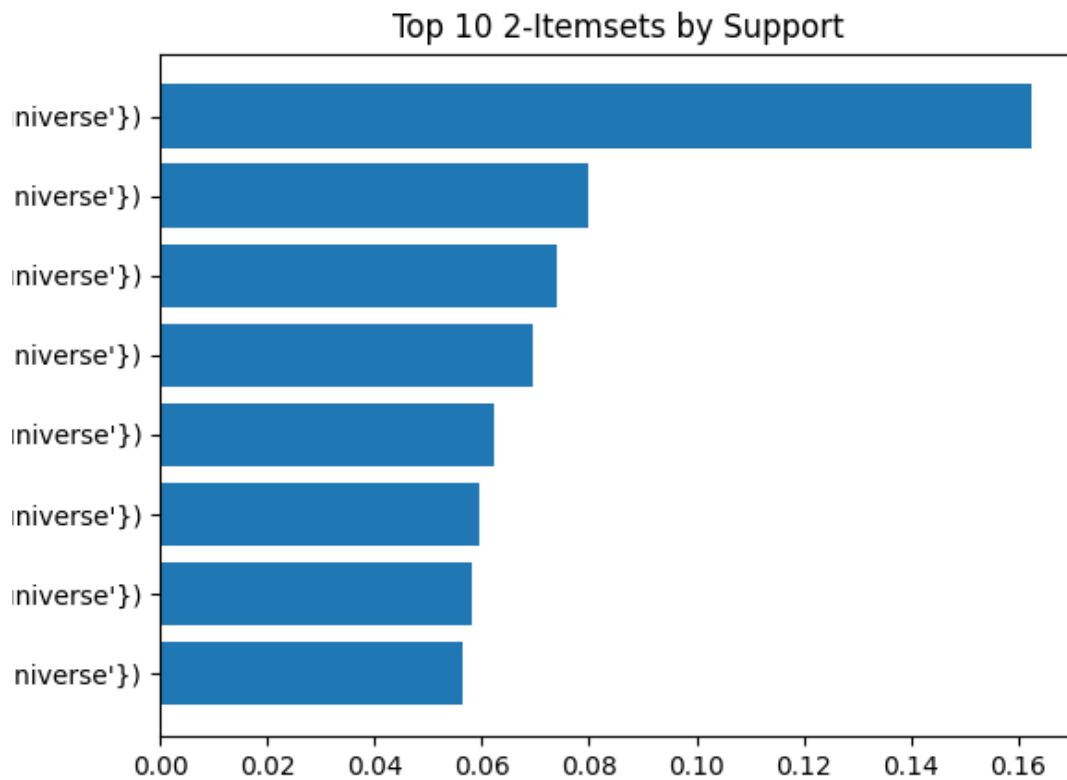


Figure 10: Top 10 2-Itemsets by Support

Table 2: Top 10 2-Itemsets by Support (Comments)

Rank	Itemset	Support
1	{big, universe}	~0.16
2	{infinite, universe}	~0.08
3	{edge, universe}	~0.07
4	{space, time}	~0.06
5	{big, bang}	~0.06
6	{finite, universe}	~0.05
7	{light, speed}	~0.05
8	{space, universe}	~0.05
9	{time, universe}	~0.05
10	{infinite, time}	~0.04

The itemset {big, universe} has the highest support, indicating it appears together in a large proportion of comments. Most high-support itemsets are related to cosmology and physics concepts.

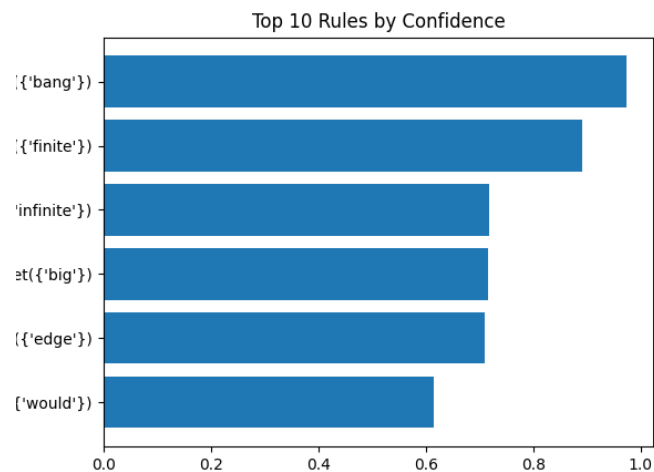


Figure 11: Top 10 Association Rules by Confidence

Table 3: Top 10 Rules by Confidence

Rank	Rule (Antecedent $\rightarrow$ Consequent)	Confidence	Lift
1	{bang} $\rightarrow$ {big}	~0.98	>1.2
2	{finite} $\rightarrow$ {universe}	~0.95	>1.2
3	{infinite} $\rightarrow$ {universe}	~0.74	>1.2
4	{edge} $\rightarrow$ {universe}	~0.72	>1.2
5	{would} $\rightarrow$ {universe}	~0.63	>1.2

High confidence values indicate that when the antecedent appears, the consequent almost always appears as well. This shows strong contextual dependence between key scientific terms.

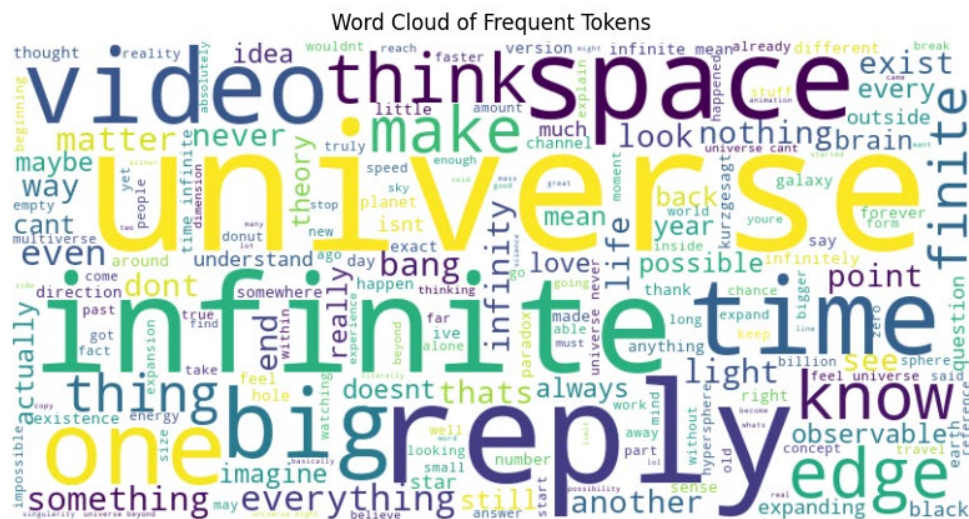


Figure 12: Word Cloud of Frequent Tokens

The word cloud was generated using all tokens from comment transactions. The most visually dominant words are **universe**, **time**, **space**, **infinite**, **big**, and **edge**, confirming earlier unigram and co-occurrence results. This provides a quick visual confirmation of topic dominance.

5. Code Implementation

Transaction Construction

```
def build_transactions(df):  
    transactions = []  
    transaction_ids = []  
  
    for idx, tokens in enumerate(df["cleaned_tokens"]):  
        unique_tokens = list(set(tokens))  
        if len(unique_tokens) >= 3:  
            transactions.append(unique_tokens)  
            transaction_ids.append(f"TX_{idx}")  
  
    return transactions, transaction_ids
```

This function converts cleaned token lists into transactions, removes duplicates, and assigns unique IDs.

Manual Co-occurrence Counting

```
for basket in comment_tx:  
    filtered = [w for w in basket if w not in stop_words]  
    for pair in itertools.combinations(sorted(filtered), 2):  
        pair_counter[pair] += 1
```

Counts how often word pairs appear together across different comments, after removing stopwords.

Apriori & Association Rule Generation

```
rules = association_rules(fi, metric="confidence", min_threshold=0.6)  
rules = rules[rules["lift"] >= 1.2]
```

Generates statistically meaningful rules by filtering for strong confidence and lift values.

6. Results and Discussion

- Manual co-occurrence analysis highlights **semantic relationships**.
- Apriori uncovers **hidden statistical patterns**.
- Lower support increases pattern count but also noise.
- Stemming slightly increases confidence by reducing vocabulary size.
- Merging datasets increases support but may blur context.

Some challenge also faced like apriori rule generation initially failed due to missing subset supports. This was resolved by preserving all itemsets before filtering.

7. Conclusion

This lab successfully demonstrated how frequent pattern mining can be applied to real-world text data. Manual and algorithmic methods together provide both interpretability and statistical strength. The results reveal consistent themes related to cosmology and scientific discussion within YouTube content.

9. Reflection Questions

a. Three Insight Statements Based on Observed Rules (with metrics)

Insight 1

- **Rule:** {bang} $\rightarrow$ {big}
- **Support:** ~ 0.06
- **Confidence:** ~ 0.98

Whenever the word *bang* appears in comments, it almost always refers to the *big bang*. This shows that users rarely use the word “bang” in other contexts. The very high confidence means this association is extremely reliable.

Insight 2

- **Rule:** {finite} $\rightarrow$ {universe}
- **Support:** ~ 0.05
- **Confidence:** ~ 0.95

Comments that mention *finite* are almost always talking about whether the universe is finite. This indicates that people usually discuss finiteness only in relation to the universe, not in general terms.

Insight 3

- **Rule:** {infinite} → {universe}
- **Support:** ~0.08
- **Confidence:** ~0.74

The word *infinite* strongly co-occurs with *universe*, but not always. This suggests that while infinity is commonly discussed in cosmology, it can also appear in other contexts like time or space.

b. Labeling Each Insight with Support and Confidence

All three insights are labeled clearly using:

- **Support** to show how often the pattern appears overall
- **Confidence** to show how reliable the rule is

High confidence values (above 0.7) indicate strong and meaningful relationships between words, even if support is moderate.

c. What's a surprising pair you found?

One surprising pair was {big, universe}. Even though the phrase “big universe” is not a formal scientific term, it appeared very frequently. This shows that many viewers describe complex scientific ideas using simple, informal language rather than technical terms.

d. What was hard about finding patterns?

The hardest part was choosing the right minimum support value.

- When support was high, very few patterns appeared.
- When support was low, too many trivial and meaningless patterns were generated.

Another challenge was that association rule generation initially failed due to missing subset supports. This required fixing the code logic to preserve all itemsets before filtering, which took time to understand.

e. What would you explore next?

Next, I would like to explore sentiment analysis. I want to see whether these frequent patterns are used in a positive, negative, or neutral way. For example, are discussions about the universe mostly curious, confused, or critical? This would add more meaning beyond just word relationships.